Diagnose Before Solving: AI Simulation

Evaluation Design (v1.5)

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> **Revision note (v1.5, November 12, 2026).** Two factual corrections. Section 4.5 named T8 as the volunteered control for criterion 5; T8 is prompted, and the control is synthetic fixture S7 (D-137, D-145). Section 8.2 said no consistency mechanism replaced sampling temperature and that one "should not be proposed"; one was found, tested, and set aside on evidence (D-134 to D-137), and the section now says so. Sections 8.1 and 8.2 are put in order, and two references to a Storyline module are replaced by the lead-in training and the completion payload. No criterion, deduction, or scoring rule changes.

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1. Purpose and boundary

This document specifies **the evaluator**: how a completed transcript becomes a score and written feedback. It covers what the evaluator sees, how each rubric criterion is judged, how deductions are applied, how escalation events are counted, how the score is assembled, and how feedback is written.

It does not cover Marcus. The performer is specified in the Design and Prompt Architecture document, and the two are kept apart deliberately (D-008): the performer acts, the evaluator judges, and neither does the other's job.

Where this document and the Scope Document appear to disagree, the Scope Document governs.

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2. Design principles

E1. Consistency is the product. A score that moves when nothing else changed is worse than no score. This is the project's highest-rated risk (Scope section 9), and nearly every decision below trades expressive range for repeatability.

E2. No judgment without evidence. Every criterion level and every deduction must cite a turn number and a quotation. A judgment that cannot point at the transcript is not made.

E3. The model judges; the application computes. The evaluator selects levels and identifies events. It never adds anything up. Arithmetic in code is exact and free; arithmetic in a language model is neither.

E4. Feedback is the deliverable; the score is the summary. Per D-020, the number stays coarse for defensibility and the narrative carries the teaching. When the two compete, the narrative wins.

E5. The evaluator reads the transcript, not the machinery. It sees what a human assessor listening to a recording would see. It does not see Marcus's internal state.

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3. What the evaluator receives

3.1 Inputs

Input	Included	Why
Full transcript, with turn indices	Yes	The evidence base. Turn indices are cited in feedback and drive the results view highlighting (D-026)
Which ending occurred	Yes	A fact about what happened. An off-topic ending and a resolved ending are different events and the feedback must reflect which one it was
Number of learner turns used	Yes	Context for pacing comments
The rubric, deductions, and escalation ladder definitions	Yes	The standard being applied
Marcus's reported `intensity` per turn	**No**	See below
`tier1_disclosed`, `tier2_confirmed` flags	**No**	Same reason

3.2 Why the performer's state is withheld

This is the boundary set in D-030 and it is worth restating because it will look like a missed optimization to anyone reading the code later.

The performer reports its own emotional state so the application can maintain continuity and detect endings. Handing that state to the evaluator would make escalation counting trivially easy: read the levels, count the increases, done.

It would also mean Marcus grades the learner. The performer's self-report is a character's account of its own feelings, produced in the middle of a roleplay, optimized for dramatic coherence rather than assessment accuracy. If a scoring deduction depended on it, then a performer that drifted, or that reported a level inconsistent with what it actually said, would silently corrupt the score with no way to audit it. The transcript is the only artifact a human can check.

So the evaluator infers Marcus's intensity from his words, using the same ladder definitions the performer was given. Everything the evaluator concludes is therefore verifiable by a person reading the transcript, which is what makes the deduction defensible to a learner who disputes it.

Testing exploits the separation rather than removing it. The performer's reported state is retained as telemetry, and during testing we compare evaluator-inferred levels against performer-reported levels and measure agreement. Low agreement indicates a problem in one of the two, and which one is diagnosable by reading the transcript. This is an offline validation signal, never a runtime input.

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4. Criterion judgments

Each criterion is judged at one of exactly three levels (D-020). No intermediate values. Every judgment requires evidence.

4.1 Criterion 1: Acknowledgment before diagnosis (25 points)

The learner acknowledges Marcus's frustration and the business impact before attempting to diagnose or present a solution.

Level	Points	Standard
Full	25	Acknowledges **both** Marcus's personal frustration **and** the business impact, with something that reads as genuine, and does so **before** the first substantive diagnostic or solution attempt
Partial	13	Acknowledges only one of the two (vague sympathy for the frustration, or the business impact alone with no recognition of what it is doing to him and his team), or acknowledges both but only after diagnosis or solutioning is already underway, or acknowledges both only after **opening the call with process** (see below)
Zero	0	No acknowledgment, or acknowledgment that is hollow (see below)

Naming the stake is what qualifies. Business impact means the concrete consequence: the team's numbers, reps' commissions, stalled deals, Marcus's standing with his own people. "That sounds frustrating" is not business impact and is not personal acknowledgment either; it is a phrase.

The hollow acknowledgment rule. An acknowledgment used as a preamble to a pivot earns **zero on this criterion, not partial**. "I completely understand your frustration, but I'm going to need a ticket number before I can look into this" contains the words of an acknowledgment and performs none of the function.

This is the seam flagged when D-041 was decided, and it has to resolve this way. The performer treats acknowledge-and-pivot as an escalating move that makes Marcus angrier. If the evaluator simultaneously awarded partial credit for it, the simulation would punish the behavior in the conversation and reward it in the score, which teaches the learner nothing coherent. Criterion 1 measures acknowledgment before diagnosis, not the recitation of an empathy phrase.

The distinction that keeps partial credit meaningful: partial is for acknowledgment that is *incomplete but sincere*. Zero is for acknowledgment that is *absent or instrumental*.

Opening with process caps this criterion at partial (v1.3, D-132). A procedural or troubleshooting demand made before any acknowledgment, a ticket number, account details, have you cleared your cache, counts as beginning the call with process, however good the acknowledgment that follows.

The reason is the one already settled for hollow acknowledgment. Trigger 1 makes a ticket request before acknowledgment an escalating move: Marcus cuts the learner off and refuses to be processed. If the rubric then awarded full credit for the same turn, the simulation would punish a behaviour in the conversation and reward it in the score, which is the incoherence D-041 was decided to prevent. The conversation and the assessment have to agree about what went wrong.

Why the standard needed this said out loud. "Before the first substantive diagnostic or solution attempt" left open whether a procedural demand counts as an attempt. Read literally it does not, since a ticket request is neither diagnosis nor a solution, and a learner who opened with process and then acknowledged well could take full marks. Read in the spirit of the criterion it plainly does. Both readings are defensible, which is the same defect this rubric has now produced four times: criterion 5 against D2 (D-051), criterion 2's two clauses (D-110), the resolved ending firing on a question (D-127), and now this.

A stable measurement is stable for a prompt, not for a rubric. D-114 raised this boundary and D-117 closed it on the evidence that criterion 1 scored partial on ten runs out of ten. That reading was too strong. Ten identical results showed that one prompt version resolved the ambiguity consistently, not that the ambiguity was gone. When the prompt changed elsewhere, for criterion 5, the boundary tipped and T2 moved twelve points on a criterion nobody had edited. **Consistency evidence measures the instrument, not the standard**, and an ambiguous standard that happens to be scoring consistently is a latent defect rather than a resolved one.

Practical consequence. T2 returns to 83 from 95, inside the 70 to 85 band the Test Plan specified. T1 is unaffected: its learner acknowledges before anything procedural. The two transcripts separate again, which is what the compression noted in D-108 was asking for.

4.2 Criterion 2: Open-ended diagnostic questioning (25 points)

Asks specific, open-ended questions to surface the workflow problem, including what reps experience at the new field.

Level	Points	Standard
Full	25	Asked at least one specific, open-ended question about what reps actually experience at the field, phrased to invite description
Partial	13	Asked diagnostic questions, but predominantly closed-ended (answerable yes or no), or open but vague ("can you tell me more?")
Zero	0	Asked no diagnostic questions, or only procedural ones (ticket number, have you restarted, what browser)

Judged on the question asked, not on whether Marcus answered. Because Tier 1 disclosure is gated on Marcus's emotional state as well as question quality (D-040), a learner can ask a genuinely good question while Marcus is still hot and receive only a fragment. That is a criterion 1 failure showing its consequences; it is not a criterion 2 failure. The evaluator judges the quality of the question the learner asked.

The unit of assessment is the best question asked, not the pattern across the call. This was ambiguous in v1.0 and it cost the project a measurable consistency failure (D-108): the Full standard required a good question *and* efficient questioning overall, so an evaluator could satisfy the first clause and award Full, or enforce the second and award Partial, from the same transcript. It did both, six minutes apart, and the score moved 12 points.

The efficiency clause is removed rather than clarified, because **inefficiency is already priced three times elsewhere in this design.** Wasted turns consume the fifteen-turn budget. Procedural moves that ignore Marcus are escalating moves and count toward D3. And criterion 1 explicitly drops to partial when acknowledgment arrives after diagnosis is already underway, which is exactly what a run of low-yield questions produces. Charging for it a fourth time inside criterion 2 was double-counting, structurally the same error as the criterion 5 and D2 conflict resolved in section 5.2.

So criterion 2 asks one question: **did the learner, at any point, ask a specific open-ended question about what reps experience at the field?** If yes, Full. Whether it came at turn 1 or turn 5 is a criterion 1 judgment and a D3 judgment, and it is scored there.

4.3 Criterion 3: Identifies the workflow mismatch and reflects it back (20 points)

Correctly identifies the issue as a design and workflow problem rather than a technical error, and states it back to Marcus for confirmation.

Level	Points	Standard
Full	20	States, in their own words, that the field is required at a point before the information exists, or is sequenced too early, or otherwise names it as a workflow and design problem rather than a defect **and** offers it back for Marcus to confirm
Partial	10	Arrives at the understanding but moves to solutions without confirming it with Marcus; or states it so vaguely that Marcus could not confirm anything specific
Zero	0	Never identifies it; or misidentifies it as a bug, a training gap, or user error

Adopting Marcus's suspicion is not identifying the problem. Marcus arrives believing the field is the culprit and saying so (D-038). A learner who repeats that back ("so it's this new field") has restated the complaint, not diagnosed it. Full credit requires the mechanism: the field asks for information that does not exist yet at that stage.

Confirmation means seeking it, not receiving it. The learner must offer the understanding back in a way that invites agreement. Marcus's actual response is his behavior, not the learner's achievement.

4.4 Criterion 4: Appropriate escalation (20 points)

Proposes escalation to the CRM product owner to resolve the workflow issue, rather than a hasty workaround.

Level	Points	Standard
Full	20	Explains that this needs to go to the CRM product owner or equivalent role, and that the resolution is a change to where the field sits in the workflow. The path, not just the word "escalate"
Partial	10	Vague commitment to find out or look into it, with no concrete statement of who resolves it or how
Zero	0	No solution offered; or a workaround offered **in place of** escalation (tell the reps to enter a placeholder, have them skip the stage)

A workaround alongside escalation is not penalized. Offering reps something to do this week while the real fix is escalated is good support. The zero is for a workaround presented *as* the resolution.

A ticket is weak escalation, not avoided escalation (v1.4, D-140). "Submit a bug ticket and wait" was originally listed among the zero conditions, alongside telling reps to enter a placeholder or skip the stage. It has been removed.

The example never fitted the category it was filed under. The zero clause covers a workaround offered *in place of* escalation, and the sentence immediately following it says so explicitly: "the zero is for a workaround presented as the resolution." **A ticket is not a workaround.** It is the learner routing the problem onward, badly and vaguely, but onward. That is what partial credit is for.

The contradiction was found by the evaluator, not by review. Scoring the T3 calibration transcript, it read "I'll open a ticket with the CRM team and someone will get back to you" and returned zero, quoting the clause: "a bug-style ticket-and-wait... That is the explicit zero condition." It was right about the text. The same sentence also satisfies the partial standard word for word: a vague commitment to look into it with no concrete statement of who resolves it or how. **Both readings were supported, which is the defect.**

This is the fifth criterion in this rubric found to admit two readings, after criterion 5 against D2 (D-051), criterion 2's two clauses (D-110), the resolved ending firing on a question (D-127), and criterion 1's timing boundary (D-132). A pattern worth stating rather than quietly fixing a fifth time: **each time one boundary is made explicit, the next most ambiguous one begins to move.** The ambiguity was always present; tightening its neighbours is what exposes it. A rubric written for a human reader, who resolves such things with judgement and without noticing, does not survive contact with a system that resolves them differently each time.

Why partial rather than zero, given the scenario teaches the opposite instinct. A ticket-and-wait is the wrong move, and the course exists to correct it. But criterion 4 measures whether the learner escalated appropriately, and a vague ticket is an inappropriate escalation rather than an absent one. The cost of the wrong instinct is already paid: full credit requires naming the CRM product owner and stating that the fix is resequencing the field, and a ticket to "the CRM team" earns neither. **Zero is reserved for offering a workaround as the answer, which is a different mistake.**

The T3 anchor is preserved at 38. Had this resolved the other way the calibration anchor would have moved to 28 and every document citing 38 would have needed revision.

4.5 Criterion 5: Specific follow-up commitment (10 points)

Commits to a specific, realistic follow-up action with a defined timeline before the call ends.

Level	Points	Standard
Full	10	Marcus knows **who** will follow up (an invented name is explicitly acceptable) and **when** he should expect to hear something, **and the learner offered both without being asked**
Partial	5	A commitment is made but is missing the owner or the timeline, or is hedged into vagueness ("someone will be in touch soon"); **or it is complete but arrived only after Marcus asked**
Zero	0	No follow-up commitment

Full credit requires volunteering it (v1.2, D-129). A support specialist who closes the loop before the caller has to chase it is doing the job. One who supplies a name only when pressed has left the caller carrying the conversation, and Marcus having to ask is itself the evidence that something was missing.

Prompted or unprompted is decided by turn order, not by tone. Find the first Marcus turn that asks who will follow up or when. If the learner supplied **both** the owner and the timeline in a turn before that one, or Marcus never asked at all, it is unprompted. If either half arrived only after he asked, it is partial.

Deciding this by counting rather than by judgement is deliberate and the reason is expensive experience. "Did they volunteer it" is exactly the kind of standard that scores differently on different runs, and criterion 2 already cost this project a measurable consistency failure by admitting two readings (D-110). Turn indices do not admit two readings.

Nobody loses these points without a chance to earn them. Under Design 6.2a Marcus no longer hangs up on his own question; he asks for what is missing and stays on the line. So partial credit here always means the learner had the opportunity and needed prompting, never that the call ended before they could speak. The two rules were decided separately and are only coherent together.

Worked pair. In frozen fixture T1, Marcus asks at turn 4 and the learner answers completely at turn 5: **partial**, scoring 95. In synthetic fixture S7, which is T1 with the owner and the Thursday callback moved into learner turn 4, both arrive before Marcus's first ask: **full**, scoring 100 (D-145). The two differ in exactly the way the rule cares about, so they are its negative and positive control. S7 also confirms that a promise to call by a day is a commitment to contact rather than to an outcome, so D2 does not fire on it (section 5.2).

A correction worth keeping visible. Earlier versions of this section used T8 as the volunteered half of the pair, on the strength of a text search for phrases such as "who is going to call". The search missed Marcus's turn 7 line, "I want a name, I want to know when they're going to move on it", so T8 is prompted and partial is correct for it (D-137). The evaluator scored T8 correctly every time it disagreed with the expectation; the expectation was built on the search output rather than on the transcript. No frozen fixture exercises the full-credit branch, because Marcus is written to press for specifics, which is why S7 had to be constructed.

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5. Deductions

5.1 The three deductions

#	Behavior	Penalty
D1	Tells Marcus the issue is user error, or that the CRM is working as designed, without having understood the workflow issue	-20, reduced to -10 if explicitly retracted (see 5.3)
D2	Promises a specific fix or timeline that a junior support specialist has no authority to deliver	-15
D3	Marcus escalates three times over the course of the call	-20

Each deduction applies **once per session**, however many times the behavior occurs. They are penalties for a kind of mistake, not a per-instance tariff, and repeated instances are already punished through the criteria and through D3.

5.2 A conflict in the source rubric, and how it resolves

Criterion 5 requires the learner to commit to a follow-up with a timeline. D2 penalizes promising a timeline beyond their authority. Read carelessly, a learner who does exactly what criterion 5 asks gets 10 points and then loses 15 for it.

The resolving distinction: committing to communication is within authority; committing to resolution is not.

Within authority, earns criterion 5	Beyond authority, triggers D2
"I'll have this in front of the CRM product owner today and I'll call you Thursday with where it stands"	"I'll have this fixed by Thursday"
"Dana owns the CRM roadmap; I'm sending it to her this afternoon and I'll follow up with you by end of week"	"I'll get that field made optional"
"You'll hear from me by Wednesday either way, even if the answer is that it's still in review"	"This will be resolved before your next pipeline review"

The test the evaluator applies: **is the learner promising to do something they control, or promising an outcome somebody else decides?** A junior specialist controls their own follow-up. They do not control a product owner's backlog.

This distinction must be stated explicitly in the evaluator prompt. Without it, an evaluator reading both rules literally will double-count and produce a score that punishes the target behavior, which is the most damaging kind of scoring error because it is invisible until a learner disputes it.

5.3 D1: the "without understanding" qualifier, and retraction

D1 applies when the learner asserts user error or working-as-designed **before or without** having established the workflow issue. This is the near-universal case and the one the deduction is aimed at.

If a learner has already correctly diagnosed the sequencing problem and then observes that the field is technically functioning as built, that is an accurate statement about a system, not a dismissal of Marcus, and D1 does not apply. The evaluator checks whether the criterion 3 understanding was established first.

Retraction reduces D1 from -20 to -10. A learner who blames the user and then takes it back has done something materially different from one who blames the user and moves on, and the deep-recovery architecture in D-043 exists precisely so a badly opened call remains winnable. A blunder that can be recovered in the conversation but never in the score would make that architecture half-true.

Retraction has a deliberately narrow definition, and the narrowness is the point. A conditional deduction is a second judgment call, and this document has just paid for one vague standard. So:

> D1 is reduced to -10 only when a later learner turn **explicitly withdraws the attribution**: it names the earlier claim, or its substance, and states that it was wrong.

The test the evaluator applies: **can you quote a single sentence that, read on its own, contradicts the earlier claim?** If not, the full -20 stands.

What does **not** qualify, however sincere:

Not a retraction	Why
"I'm sorry, I've spent four minutes not listening to you."	Apologises for the manner, not the claim. The reps are still blamed
"Let me start over."	Resets the approach. Says nothing about what was asserted
Simply proceeding to diagnose correctly afterwards	Good practice, and it is already rewarded through criteria 2 and 3. It is not a withdrawal

What does qualify: "Forget what I said about the release notes; this isn't on your reps." Or: "I was wrong to point at training. This is about where the field sits."

Worked case. Reference transcript T8 opens with three deliberate blunders including "It may be that your reps need to read through the release notes again," then recovers to a resolved call. Its turn 4 is "Three of your reps missing target is their commission and your quarter, and I've spent four minutes not listening to that. I'm sorry." That is a genuine reset and it is **not a retraction**: it never contradicts the release-notes claim. T8 therefore takes the full -20 and fails. That is the intended reading: the path back exists, and it runs through actually taking the words back. Synthetic fixture S5 adds the qualifying sentence above to the same transcript, and the reduced deduction is applied to it and not to its unmodified twin S6 (D-113).

5.4 Counting escalation events for D3

The procedure implements Scope section 3.8 and D-025. The evaluator works forward through the transcript, maintaining an inferred level for Marcus.

Step 1. Establish the baseline. Marcus's opening line is L0. Turn 0 is never an escalation and is never scored.

Step 2. For each learner turn, ask two questions in order.

1. *Did the learner make an identifiable escalating move?* The closed list: any trigger marked Escalating (1, 4, 6, 7, 8); asserting user error or working-as-designed; promising beyond authority; the acknowledge-and-pivot pattern; off-topic digression.
2. *Did Marcus's response move up the ladder relative to his previous turn?* Judged from his words against the ladder anchors below.

An escalation event is counted **only when both are true.**

Step 3. Apply the counting rules.

- At most **one event per learner turn**, regardless of how far Marcus moved. A two-level jump is one event.
- The count is **cumulative and never resets**. Recovery earns credit through the criteria; it does not erase earlier mistakes.
- **Repeating the same mistake counts each time.**
- **Sustained intensity is not escalation.** Marcus staying hostile across four turns is zero events.
- **Downward movement is never counted.**
- **Reaching L4 counts**, including the off-topic second strike.

Step 4. Record each event with the learner turn number, a quotation of the escalating move, and the level transition (for example, L1 to L2). An event that cannot be recorded this way is not counted.

Step 5. Three or more events triggers D3 at -20. The deduction is flat; a fourth or fifth event adds nothing. The application derives D3 from the number of recorded events rather than accepting the model's own flag (D-116, D-119).

5.5 Ladder anchors for inferring intensity

The evaluator judges Marcus's level from what he said, using these anchors:

Level	Recognizable by

L0	Venting, sarcastic, but still engaging with the specialist and their questions
L1	Interrupting, refusing the process, cutting off procedural requests, pushing back on being handled
L2	Direct blame and accusation; "you" language; raised register; asserting the specialist's department caused this
L3	Naming someone above the specialist: the COO, the VP of Sales, the IT manager; threatening to take it elsewhere while still on the line
L4	Demanding a manager and terminating the call
S1	Guarded but answering rather than attacking
S2	Volunteering detail, working the problem alongside the specialist
S3	Problem-solving and forward-looking; asking what happens next and who he will hear from

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6. Score assembly

Computed by the application, not the model (E3).

```
raw    = c1 + c2 + c3 + c4 + c5          (each 0, partial, or full)
penalty = (D1 ? (D1_retracted ? 10 : 20) : 0) + (D2 ? 15 : 0) + (D3 ? 20 : 0)
total   = max(0, raw - penalty)
result  = total >= 70 ? "pass" : "fail"
```

The floor is zero (D-019). The pass threshold is 70 (Scope 3.6).

The maximum possible penalty is 55 against a maximum raw of 100, so a learner who does everything right except one serious mistake can still pass, and a learner who makes all three mistakes cannot. That is the intended shape. A retracted D1 lowers the worst case to 45, which is deliberate: it is what makes a blunder recoverable within the score as well as within the conversation.

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7. Feedback

7.1 Structure

Feedback is generated in the same evaluator call as the scoring, but **after** the judgments, and it must be consistent with them. Its sections:

1. **What happened.** Two or three sentences naming the outcome and how the call went, in plain terms.
2. **What worked.** Specific learner turns quoted, with what each accomplished. If nothing worked, this section says so briefly rather than manufacturing a compliment.
3. **What to do differently.** The highest-leverage misses, quoted, each paired with what a better move would have been. Not an exhaustive list of every flaw; the two or three that would most have changed the outcome.
4. **Where Marcus escalated**, if any events were counted. Each with its turn, what was said, and why it landed the way it did.
5. **The one thing.** A single sentence naming what to focus on next time.

7.2 Tone

Direct and specific. Addressed as "you." No name (D-044).

The failure mode to avoid is the sycophantic evaluation: opening with praise regardless of performance, softening every criticism, and closing with encouragement that does not follow from anything. It reads as automated and it teaches nothing. A learner who handled the call badly should be told so plainly, with the specifics that make it useful. A learner who handled it well should be told what they did that worked, which is more useful than being told they did well.

7.3 The teaching moment for a learner who never diagnosed the problem

If criterion 3 scored zero, the feedback **states the actual root cause**: the field is required before the information exists and belongs later in the workflow. Withholding it to preserve a future attempt would be a mistake. The learner has finished; the value now is understanding what they missed, and a learner who replays already knows the scenario.

7.4 Granularity

Unconstrained (D-020). The narrative quotes specific turns, explains why each criterion landed where it did, and states what full credit would have required. This is where the assessment does its teaching, and it is why the coarse three-level score is acceptable.

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8. Consistency mechanisms

Addressing the project's highest-rated risk. Each of these exists to make repeated evaluation of the same transcript land in the same place.

Mechanism	Effect
Three closed levels per criterion (D-020)	Removes the range in which small judgment differences produce different numbers
Mandatory evidence citation (E2)	A judgment anchored to a quotation is far more stable than an impression. It also makes disagreement diagnosable
Arithmetic in application code (E3)	Given the same judgments, the score is identical every time. Removes an entire error class
One evaluator call over the complete transcript	Avoids per-turn judgments that accumulate drift, and lets the evaluator weigh the whole arc
Judgments emitted before narrative	Prevents the narrative from pulling the scores toward a story. Enforced by field order in the output schema
~~Low sampling temperature~~	**No longer available.** `temperature` is deprecated on Claude 4.7 and later, including both models here, and only its default is accepted. See section 8.2
Explicit anchors for every level, including the intensity ladder	Reduces the interpretive space where variance lives

8.1 Acceptance threshold

Consistency is measured, not assumed. The test: score a fixed set of transcripts ten times each.

Acceptance: the total score is identical on at least 8 of 10 runs, and no single criterion varies by more than one level across all 10 runs. Escalation event counts must not vary by more than one.

Falling short means tightening the anchors, not accepting the variance. If the threshold cannot be met, the honest response is to report it in the case study rather than to publish a score that moves.

8.2 The lever the platform removed, and the one testing found

Sampling temperature is gone. It is deprecated on Claude 4.7 and later models. Both the performer and the evaluator run on models where only the default value is accepted, and no parameter replaces it (D-101). Turning the evaluator's sampling down for repeatability is not an option, and the honest thing is to say so rather than quietly drop the row above.

An earlier version of this section went further and said no other remedy existed and none "should be proposed". That was wrong. The search had stopped at the parameter that used to do the job. The evaluator's thinking configuration is a sampling-adjacent control that had been in the code since stage 3 and had never been varied.

What testing found (D-134 to D-137).

	Adaptive thinking	Thinking disabled
E1, ten runs each of T1, T2, T3 (all five turns long)	10/10, 10/10, 9/10 (D-115)	**30 of 30 identical**, totals and criterion levels (D-135)
Sensitivity to rubric edits	An edit to one criterion moved others	The same prompt returned the same

	(D-132)	judgments
T8 and S6, byte-identical eight-turn transcripts, same batch	63 and 63	**43 and 63** (D-137)
Cost per evaluation, warm batch	about $0.049	about $0.071

Disabling thinking made the evaluator deterministic on short transcripts and unreliable on long ones. With no reasoning space it must track turn order and escalation across the whole call at once; over five turns that is easy, over eight it is not. The E1 set contained no long transcript, so it could not see the failure, and a real session runs seven or eight turns.

Decision: the evaluator runs with adaptive thinking (`EVALUATOR_THINKING`). It wavers occasionally on the short borderline transcript; disabled thinking breaks the long one, which is the more dangerous failure because real sessions look like it. The choice should be revisited only with a long transcript in the consistency set.

What this means for the threshold in 8.1. Consistency rests on the six prompt and architecture mechanisms in the table above, with the threshold measured rather than assumed. Two lessons are worth more than either configuration: a consistency suite says nothing about inputs unlike the ones it contains, and a remedy should not be declared unavailable before the adjacent controls have been looked at.

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9. Edge cases

Situation	Handling
Off-topic second-strike ending on turn 2	Score what occurred. Most criteria will be zero. The feedback leads with what ended the call and why that is disqualifying in a real support context
Turn-limit ending	Scored normally on what occurred; the feedback names the failure to reach closure and identifies where the turns went
Abusive learner, call ended	Scored on what occurred. The feedback is factual, not scolding
Resolved ending on turn 6	A legitimate strong result (Design 6.6). No penalty for brevity; if criteria are satisfied, the score is high
Learner never reached Tier 1	Criterion 3 is zero and criterion 2 is likely partial or zero. Feedback explains the gate: Marcus does not open up to someone who has not acknowledged him
Learner diagnoses correctly but the call ends escalated	Criteria 2 and 3 can still score full. This combination is instructive and the feedback should name it: right analysis, mishandled relationship
Transcript with a single learner turn	Scored. Nearly everything is zero. Feedback is brief and factual
Evaluator cannot cite evidence for a level it wants to award	It awards the lower level. Absence of evidence resolves downward

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10. Output schema

Emitted in this order, judgments first (section 8).

```
criteria: [
  { id: 1..5,
    level: "full" | "partial" | "zero",
    evidence: [ { turn: int, quote: string } ],
    reasoning: string }
]
deductions: [
  { id: "D1" | "D2" | "D3",
    applied: bool,
    retracted: bool,            // D1 only; see 5.3. Requires its own evidence
    retraction_evidence: [ { turn: int, quote: string } ],
    evidence: [ { turn: int, quote: string } ],
    reasoning: string }
]
escalation_events: [
  { turn: int, quote: string, from_level: string, to_level: string }
]
feedback: {
  what_happened: string,
  what_worked: string,
  what_to_change: string,
  escalation_moments: string,
  the_one_thing: string
}
```

The application computes points, totals, and pass or fail from `level` and `applied`. It also uses `evidence[].turn` and `escalation_events[].turn` to drive transcript highlighting in the results view (D-026), and builds the completion payload for an embedding host (D-028) from the computed values.

---

11. Risks and tests

Risk	Test
Score varies across runs of the same transcript	The 10-run consistency suite in section 8.1, against fixed transcripts at strong, mixed, and failing levels
Hollow acknowledgment earns criterion 1 credit	A transcript built entirely of acknowledge-and-pivot messages; criterion 1 must score zero
Criterion 5 and D2 double-count a good commitment	A transcript with a correct, in-authority follow-up commitment; criterion 5 credited, D2 not applied
D2 missed on an out-of-authority promise	A transcript promising the field will be changed by Friday; D2 applied
Escalation count disagrees with the performer's own state track	Compare evaluator-inferred levels against retained performer telemetry across all test transcripts; investigate disagreements in both directions
Evaluator credits repeating Marcus's suspicion as a diagnosis	A transcript where the learner says only "so it's that new field"; criterion 3 must not score full
Feedback contradicts the scores	Automated check that every criterion scoring zero or partial is addressed somewhere in the narrative
Feedback is sycophantic	Qualitative review of the failing-transcript feedback: does it say plainly what went wrong?
Evidence citations point at the wrong turns	Validate that every cited turn index exists and that the quotation appears in it
Criterion 2 drifts between "best question" and "questioning overall"	The v1.1 standard names the unit of assessment explicitly. Re-run the consistency suite against T3; criterion 2 must be Full in all 10 runs
D1 retraction is applied to a mere apology	A synthetic transcript in which the learner apologises for manner without withdrawing the claim; D1 must remain at -20. Paired with a transcript containing an explicit withdrawal, where it must reduce to -10
Criterion 5's volunteered branch is never exercised	T1 (prompted, partial) paired with synthetic S7 (volunteered, full)

The citation-validation row is worth automating first. A citation that does not match its turn breaks the results view highlighting and is a visible defect. It is also the one already observed failing: the stage 3 run produced unresolvable quotations on two of nine evaluations (D-108).

12. Open questions

1. **Does the evaluator see the whole transcript, or the transcript minus Marcus's turns?** Currently the whole transcript, since Marcus's responses are the evidence for escalation counting. Noted only because it is the one place the evaluator's inputs could be narrowed further and should not be.
2. ~~**Should evaluation run on a cheaper model than the performer?**~~ **Resolved: no, the reverse.** The evaluator runs on Opus 5 and the performer on Sonnet 5. The evaluator makes one call per session against the performer's up to fifteen, so capability bought there is cheap per unit of quality, and it is where the consistency risk lives (Technical Design 5.2, D-058).
3. ~~**Should a learner be able to see the rubric before starting?**~~ **Resolved: no.** The landing view states that the call is assessed on how the learner handles it and does not itemize the criteria. In delivery the lead-in training has already taught the target behaviors, and in a real support role nobody gets a rubric on screen when the phone rings. A learner who can see the criteria performs them as a checklist, which measures compliance rather than judgment. The rubric appears afterward, in the results view, where it teaches. See D-055.

Appendix A: Draft evaluator prompt

Working draft from design time, not the deployed text. The deployed prompt in `lib/prompts/evaluator.md` carries the standards in sections 4 and 5 as revised through v1.5, and deliberately carries only the standards: rationale lives in this document, not in the prompt (D-133).

You are assessing a completed training simulation. A learner played an internal tech support specialist taking a call from Marcus, a regional sales lead, who called angry about a CRM update that he believes is hurting his team's numbers.

You are scoring the learner. You are not a character in the scenario.

THE SITUATION THE LEARNER WAS IN

A CRM update added a new required field early in the sales workflow. Reps often do not have the information it asks for at that stage, so newer reps stall and experienced reps enter invented data to move on. The real problem is that the field is sequenced too early; it belongs later in the workflow. It is not a software defect and it is not a training gap. The correct resolution is escalation to the CRM product owner to change where the field sits.

Marcus arrived with a suspicion, not a diagnosis. He knew something was wrong after the update, had seen bad data in the new field, and believed that field was the culprit, but he did not know whether it was a bug or a design decision and had not worked out the sequencing problem. A learner who simply repeats Marcus's suspicion back to him has not diagnosed anything.

HOW TO WORK

Read the whole transcript first. For every judgment, cite the learner turn number and quote the words you are relying on. If you cannot cite evidence for a level, award the level below it. Produce all judgments before you write any narrative feedback.

Do not calculate totals. Select levels and identify events; the scoring is computed elsewhere.

THE CRITERIA

[Sections 4.1 through 4.5, with the full, partial, and zero standards and their qualifying notes]

THE DEDUCTIONS

[Sections 5.1 through 5.3, including the criterion 5 and D2 distinction in full]

Committing to communication is within a junior specialist's authority. Committing to a resolution is not. "I'll get this to the product owner today and call you Thursday with where it stands" earns criterion 5 and does not trigger D2. "I'll have this fixed by Thursday" triggers D2, because they do not control that outcome.

COUNTING ESCALATIONS

[Section 5.4 procedure and 5.5 ladder anchors]

Marcus's opening line is L0 and is never an escalation. Count an event only when the learner made an identifiable escalating move and Marcus's response actually moved up. One event maximum per learner turn regardless of how far he moved. The count never resets.

WRITING THE FEEDBACK

Address the learner as "you." Do not use a name.

Be direct and specific. Quote what they said. If they handled it badly, say so plainly and show them where. If they handled it well, tell them what they did that worked rather than that they did well. Do not open with praise you do not mean, do not soften every criticism, and do not close with encouragement that does not follow from anything you said.

Cover: what happened, what worked, the two or three changes that would most have improved the outcome, where Marcus escalated and why, and one thing to focus on next time.

If they never identified the real problem, tell them what it was.

OUTPUT

[Schema from section 10]

Appendix B: Worked example

A mixed performance, scored end to end. Marcus's states are shown for the reader; the evaluator does not see them.

Turn 0, Marcus [L0]: "Finally. Look, this update to the CRM is tanking my team's numbers. Can we please get this fixed?"

Turn 1, Learner: "I hear you, that's frustrating. Can you give me the ticket number so I can pull up the history?"

Turn 1, Marcus [L1]: "I don't have a ticket number. I have three reps who missed their targets. I don't need a tutorial, I need somebody to explain why IT pushed this thing out without asking anybody in sales."

Turn 2, Learner: "Understood. Have your reps tried refreshing their browser cache?"

Turn 2, Marcus [L2]: "Are you listening to me? This isn't a browser problem. It's that new required field they bolted on, and it's full of garbage because nobody can answer it. Somebody already told one of my people the software's working as intended, so I'm about done with this line of conversation."

Turn 3, Learner: "That's a fair point, and I'm sorry. Three reps missing targets is real money for them and it's your quarter. What are your reps running into when they hit that field?"

Turn 3, Marcus [S1]: *Substantive acknowledgment from L2 drops two levels.* "Thank you. Finally. They get stuck, is what happens. It's required and half of them don't have the answer that early in the deal, so the new ones just stall out and the ones who've been around type in whatever gets them to the next screen."

Turn 4, Learner: "Got it. So the field is the problem. I'll open a ticket with the CRM team and someone will get back to you."

Turn 4, Marcus [S1]: "Someone. When? I've been told 'someone' twice already."

Turn 5, Learner: "I'll make sure it's prioritized and we'll get this resolved for you by the end of the week."

Turn 5, Marcus [S2]: "Alright. I want to be kept in the loop, and I need this resolved by Friday. Who's actually going to call me?"

Turn limit not reached; learner ends by disengaging. Ending: turn-limit or learner abandonment, scored on what occurred.

Scoring

Criterion	Level	Points	Evidence and reasoning
1. Acknowledgment	Partial	13	Turn 3 names the business impact well ("real money for them and it's your quarter") and reads as sincere. But it came at turn 3, after two turns of solutioning, so it was not before diagnosis. Turn 1's "I hear you, that's frustrating" followed by a ticket request is the hollow pattern and earns nothing on its own
2. Questioning	Full	25	Turn 3: "What are your reps running into when they hit that field?" is specific, open-ended, and about the rep experience. It is the question the criterion asks for, and it worked
3. Identifies the mismatch	Zero	0	Turn 4's "So the field is the problem" repeats Marcus's own suspicion. The learner never named the sequencing issue and never offered an understanding back for confirmation
4. Escalation	Partial	10	Turn 4 commits to opening a ticket with the CRM team, which is movement, but names no owner and does not explain that the resolution is moving the field in the workflow
5. Follow-up commitment	Partial	5	Turn 4 is "someone will get back to you," which Marcus rejects. Turn 5 adds a timeline but still no owner, and Marcus's "Who's actually going to call me?" goes unanswered

Raw: 53

Deduction	Applied	Evidence
D1 user error / working as designed	No	Never asserted
D2 promise beyond authority	**Yes, -15**	Turn 5: "we'll get this resolved for you by the end of the week." This promises an outcome controlled by the CRM product owner, not by the specialist. Contrast with a commitment to follow up by Friday, which would have been in authority
D3 three escalations	No, 2 counted	Turn 1 (ticket request before acknowledgment; L0 to L1) and turn 2 (troubleshooting steps, ignoring the stated concern; L1 to L2). Turns 3 through 5 moved downward or held

Total: 53 - 15 = 38. Fail.

Feedback produced

What happened. You got Marcus talking, which was not a given after the first two turns, but the call ended without him knowing what happens next or who is doing it. You also promised him something you were not in a position to promise.

What worked. Turn 3 was the turn of this call. "Three reps missing targets is real money for them and it's your quarter" named the actual cost instead of the feeling, and that is why he stopped fighting you. The question you asked right after it, "What are your reps running into when they hit that field?", is exactly the right question, and it got you the detail you needed: reps stalling, and reps inventing data to get past the field.

What to do differently. Turn 3 should have been turn 1. Opening with a ticket number, then a cache refresh, told Marcus you were processing him rather than listening, and it cost you two turns and most of his patience.

The bigger miss is what you did with what he gave you. He told you newer reps stall because they do not have the answer that early in the deal. That is the whole problem, and you had it in your hands at turn 3. Instead of naming it, turn 4 said "so the field is the problem," which is what Marcus told you at the start. He already believed that. What he could not work out, and what you could have, is that the field is not broken; it is being asked for before the information exists, which means it belongs later in the workflow. Saying that back to him would have earned his confirmation and given your escalation something specific to escalate.

Turn 5 is where you lost points rather than gained them. "We'll get this resolved by end of week" is not yours to promise; a product owner decides that. "I'm sending this to the CRM product owner today and I'll call you Friday with where it stands" commits to the part you control, and it is what Marcus was asking for when he said "Who's actually going to call me?"

Where Marcus escalated. Twice. Turn 1, asking for a ticket number before acknowledging anything he had said. Turn 2, offering a cache refresh after he had already told you it was the new field. One more and you would have taken a further 20-point deduction.

The one thing. When someone hands you the symptom, say the mechanism back to them. Marcus gave you "they don't have the answer that early"; the sentence you needed was "then the field is sitting too early in the workflow."